ICS4U Battleship Assignment

In this assignment, you will be designing an algorithm to choose the best move in a game of battleship, which you will use to compete against each other student in the class.

Setup and Rules

You have been provided with a filename called **battleship_studentname.py** which you will change **studentname** to your name, first and/or last, no spaces.

You will change the BOT_NAME variable to your battleship bot's nickname (max 20 characters)

def place_ships(grid):

The variable **grid** is a 10 by 10 2D-list that starts as all zeroes. You will place 5 non-overlapping ships of length 5 wherever you want on the board, at least one horizontally, one vertically, and one diagonally, using for or while loops. To place the ship, change the grid element to 1.

def make_move(opp_grid, prev_moves):

In this function, you will use the information that you know about your opponent's grid, and a list of your previous moves, to decide what your next move is.

Your opponent's grid is a 10 by 10 2D-list that has elements

- 0 for unknown element
- 2 for a known miss
- 3 for a known hit

The prev_moves is a list of lists containing your previous moves. Each element of prev_moves is a list of 3 elements, the row, the column, and whether it was a hit (True) or not (False). Your most recent previous move will be at the end of the list.

Testing your code

You will refine your strategy by playing automated games within a group of 3-4 students.

Check out the **play_1v1.py** file on Google Classroom. You need to make sure that that file, your battleship file, and a classmate's battleship file are all in the same folder. Then, open the **play_1v1.py** file so that battleship_player1 and battleship_player2 are changed to match you and the classmate's battleship filename, excluding the .py extension.

Scroll down to the bottom of the file and make the necessary name changes as well. The enter condition determines whether or not you need to press enter between each turn. The display condition determines whether or not every turn is displayed.

Writeup and Progress

Journal your progress at least once per class period over the course of the project, explaining how your testing process improved your algorithm. Include the following

- Comment on how the testing process changed your initial placement of battleships.
- Have at least two commentaries on how you resolved logical errors in your code.
- Comment on the efficiency of your code. How would it scale to a million by a million grid?
- Comment on further improvements you could make if you had more time on the project.

Rubric and grading.

	Level 1/2	Level 3	Level 4-/4	Level 4+
Correctness and sophistication	Algorithm is mostly random and has limited strategy, and may make a few mistakes	Algorithm begins to strategize, such as picking neighboring elements of known hits	Algorithm uses intermediate strategies such as continuing in a line to get an entire ship	Algorithm uses advanced strategies such as probabilities and analysis of previous moves
Progress and analysis	Writeup explains algorithm but the process work is lacking	Writeup explains algorithm and testing process with but lacks daily reflection and progression	Writeup outlines progress through the project but missing some details to tell a complete story of the process.	Writeup shows consistent progress, thorough detail, with strategies driven by extensive testing and reflection